Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	McQuillen, John		
Project Title:	Study of Two Phase Gas-Liquid Flow Behavior at Reduced Gravity Conditions		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:	National Institutes of Health/National Cancer Institute	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44135	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	OTHER
Start Date:	03/01/2000	**End Date:**	09/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	note: end date moved to 9/30/04		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fabre, J. (Institut De Mecanique Des Fluides) Colin, C. (Toulouse)		
Grant/Contract No.:	IH1011306		
Performance Goal No.:	2.6		
Performance Goal Text:	Advance the state of knowledge sufficiently to allow development of reliable and efficient heat transfer technology for space and extraterrestrial operations.		

Task Description:	The proposal calls to study specifically the following: 1) The dynamics of isolated bubbles in microgravity liquid flows including the dynamics of spherical isolated bubbles in order to study their dispersion by turbulence, the interaction with the pipe wall, the behavior of the bubbles in positive or negative pressure gradient (bubbles in accelerated or decelerated flows), the dynamics of cylindrical isolated bubbles in order to analyze their velocity with respect to liquid flow regime, their deformation in accelerated/ decelerated flows (in converging or diverging channels), and bubble/bubble interaction will be analyzed. Experiments will consist of the use of PIV and LDV to study single spherical bubble and single and two cylindrical bubble behavior with respect to their influence on the turbulence of the surrounding liquid and on the wall. 2) The dynamics of bubbly and slug flow in microgravity will be analyzed especially for the role of the coalescence in the transition from bubbly to slug flow (effect of fluid properties and surfactant), to identify clusters that promote coalescence and transition the void fraction distribution in bubbly and slug flow, to measure the wall friction in bubbly flow. The objective of this study is to examine the behavior of gas- liquid flows in reduced gravity, specifically for bubble and slug flow. The purpose of this study is to achieve a better understanding-better predictability-of two-phase (gas-liquid) flow in pipes to assist in the design of space-based power and thermal management systems and of the terrestrial-based nuclear power plants and oil and natural gas pipelines.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 09/07/2020)